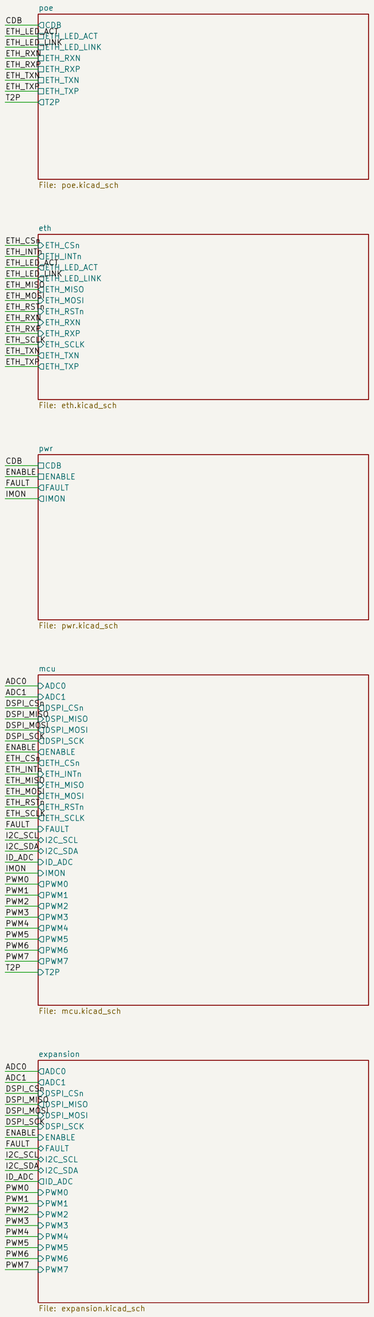
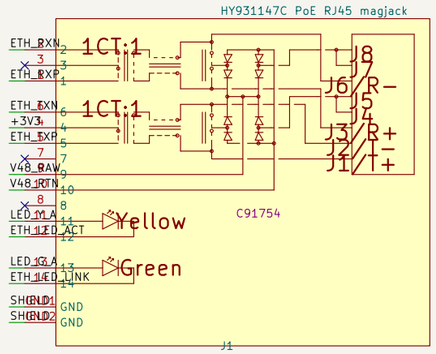


ai-ee		
Sheet: /		
File: lumina-carrier.kicad_sch		
Title: LUMINA carrier LUM-CAR-A: root		
Size: A2	Date:	**Rev: 1**
KiCad E.D.A. 10.0.3		Id: 1/6



330R 1% 0603

+3V3 — LED_Y_A

R7C23138

330R 1% 0603

+3V3 — LED_G_A

R8C23138

12.4k 1% 0805

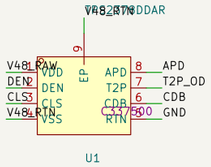
V48_RAW — DEN_TAP

R1C30908

12.4k 1% 0805

DEN_TAP — DEN

R2C30908



10k 1% 0805

+3V3 — T2P_OD

R4C17414

10k 1% 0805

T2P_OD — T2P

R5C17414

0R jumper 0603

APD — GND

R9C21189

ETH_TXP — ETH_TXP

ETH_TXN — ETH_TXN

ETH_RXP — ETH_RXP

ETH_RXN — ETH_RXN

ETH_LED_LINK — ETH_LED_LINK

ETH_LED_ACT — ETH_LED_ACT

T2P — T2P

CDB — CDB

SMBJ58A 58V 600W TVS

V48_RAW — V48_RTN

D1C2891331

90.9R 1% 0603

CLS — V48_RTN

R3C23138 (NOT FITTED — value swap, not a 2nd part)

1M 1% 0603

SHIELD — GND

R6C22935

1nF 2kV X7R 1206

SHIELD — GND

C3C9196

100nF 100V X7R 0805

V48_RAW — V48_RTN

C1C28233

10uF 100V X7R 1210

V48_RAW — GND

C2C5156756

10uF 100V X7R 1210

V48_RAW — GND

C4C5156756

10uF 100V X7R 1210

V48_RAW — GND

C5C5156756

10uF 100V X7R 1210

V48_RAW — GND

C6C5156756

PWR_FLAG

GND

GND

V48_RAW — PWR_FLAG

V48_RAW

V48_RTN — PWR_FLAG

V48_RTN

+3V3

+3V3

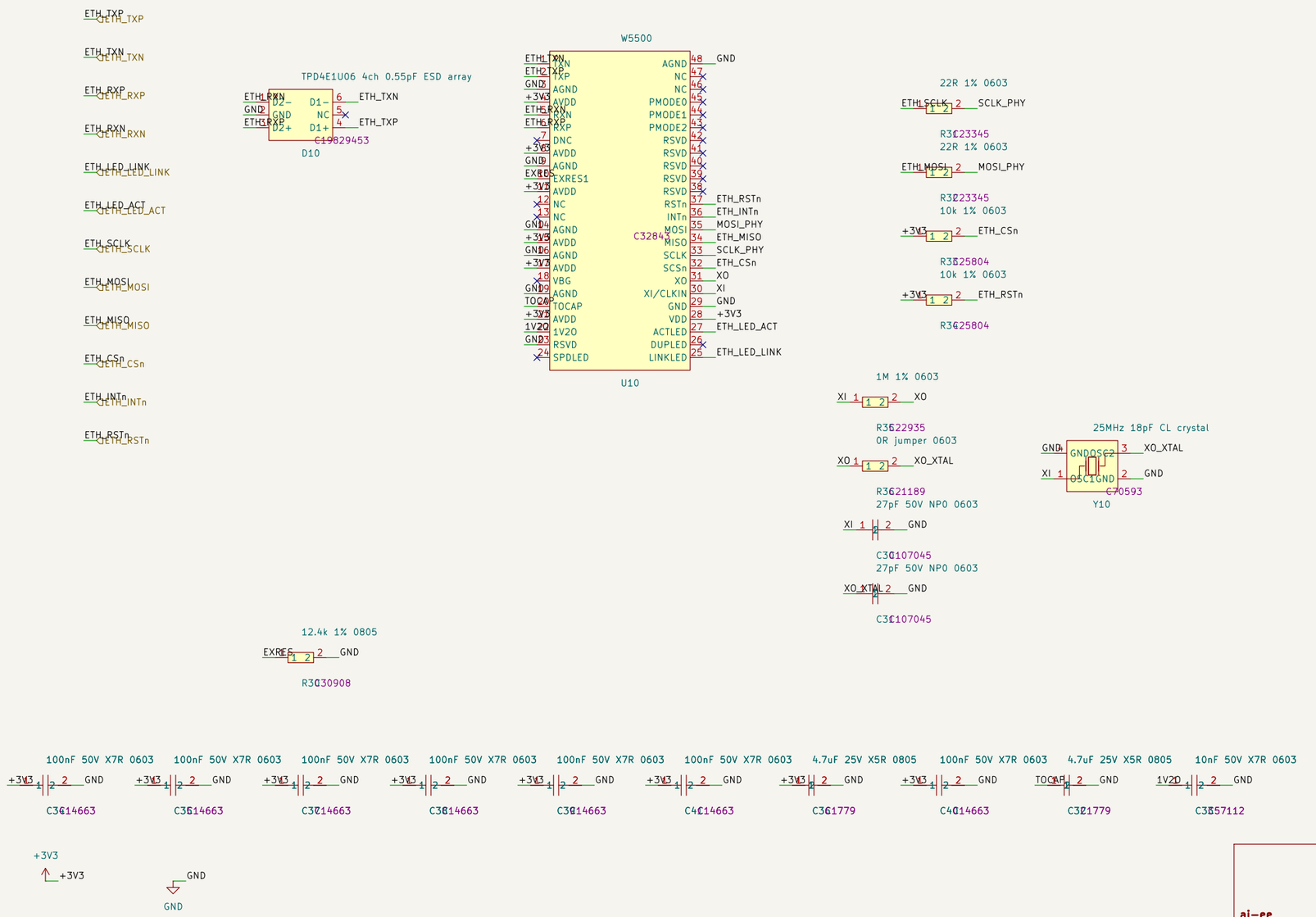
ai-ee

Sheet: /poe/

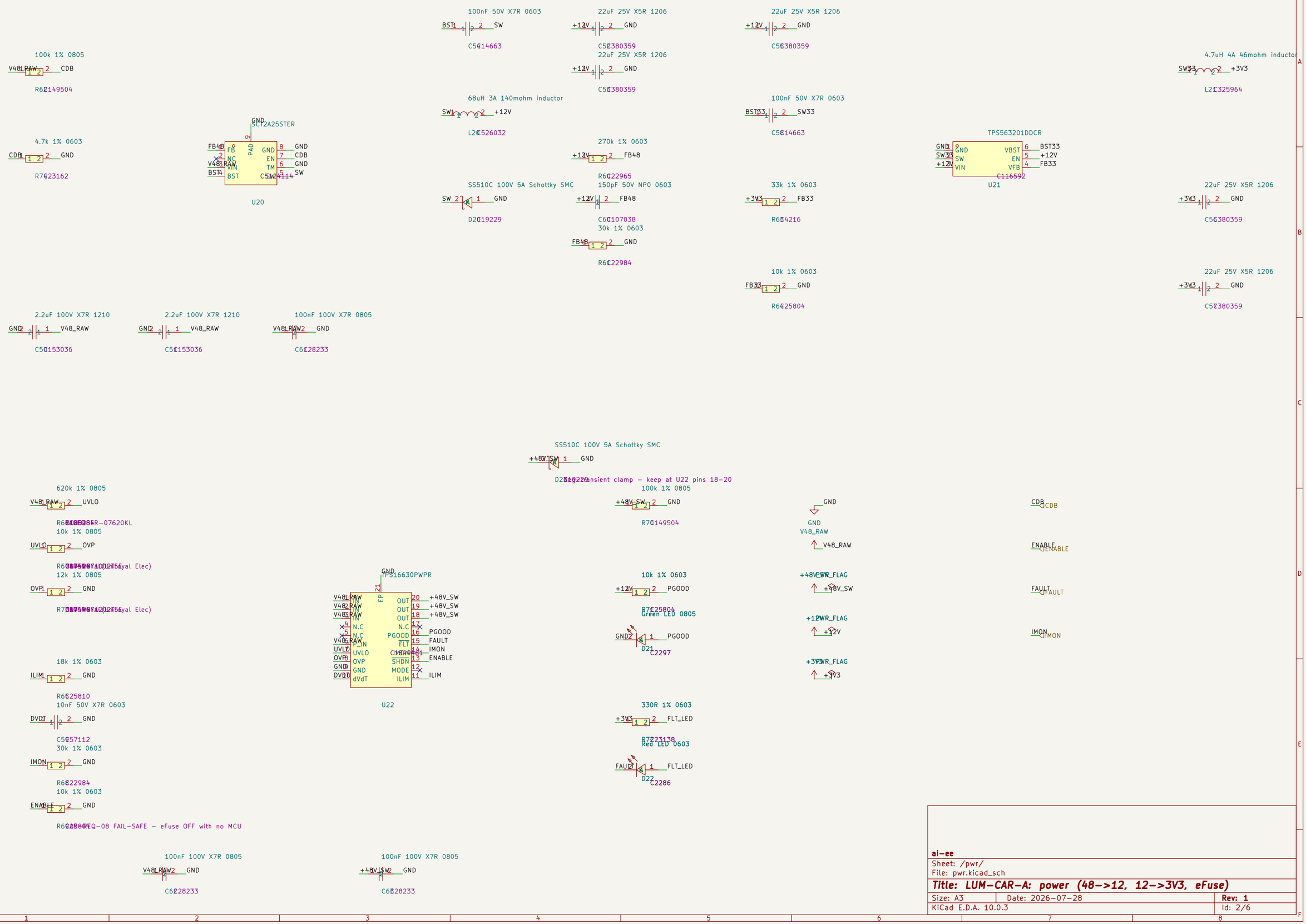
File: poe.kicad_sch

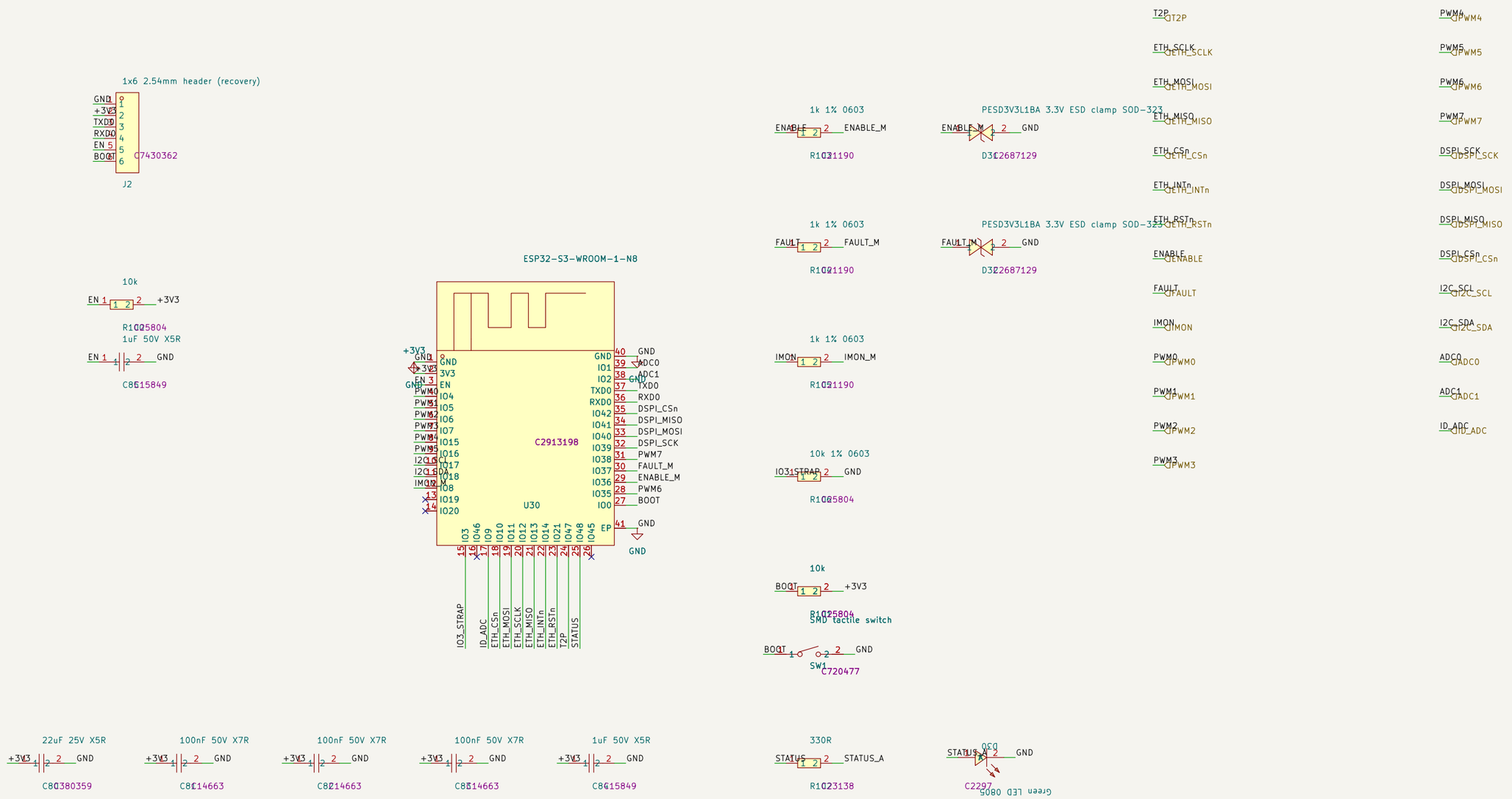
Title: LUM-CAR-A: PoE PD front end (48 V domain)

Size: A3	Date: 2026-07-28	**Rev: 1**
KiCad E.D.A. 10.0.3		Id: 2/6



ai-ee	
Sheet: /eth/	
File: eth.kicad_sch	
Title: LUM-CAR-A: eth - W5500 Ethernet controller	
Size: A3	Date: 2026-07-28	Rev: 1
KiCad E.D.A. 10.0.3	Id: 2/6


100nF 50V X7R 0603
22uF 25V X5R 1206
22uF 25V X5R 1206
100k 1% 0805
4.7uH 4A 46mohm inductor
68uH 3A 140mohm inductor
100nF 50V X7R 0603
SCT2A25STER
TPS563201DDCR
4.7k 1% 0603
270k 1% 0603
SS510C 100V 5A Schottky SMC
150pF 50V NPO 0603
33k 1% 0603
30k 1% 0603
10k 1% 0603
22uF 25V X5R 1206
22uF 25V X5R 1206
2.2uF 100V X7R 1210
2.2uF 100V X7R 1210
100nF 100V X7R 0805
SS510C 100V 5A Schottky SMC
TVS transient clamp — keep at U22 pins 18–20
100k 1% 0805
620k 1% 0805
SMBJ58CA-TR-07620KL
10k 1% 0805
12k 1% 0805
TPS16630PWPR
10k 1% 0603
Green LED 0805
18k 1% 0603
10nF 50V X7R 0603
330R 1% 0603
Red LED 0603
30k 1% 0603
10k 1% 0805
SMCJ6.0CA-08 FAIL-SAFE — eFuse OFF with no MCU
100nF 100V X7R 0805
100nF 100V X7R 0805
ai-ee
Sheet: /pwr/
File: pwr.kicad_sch
Title: LUM-CAR-A: power (48->12, 12->3V3, eFuse)
Size: A3
Date: 2026-07-28
Rev: 1
KiCad E.D.A. 10.0.3
Id: 2/6



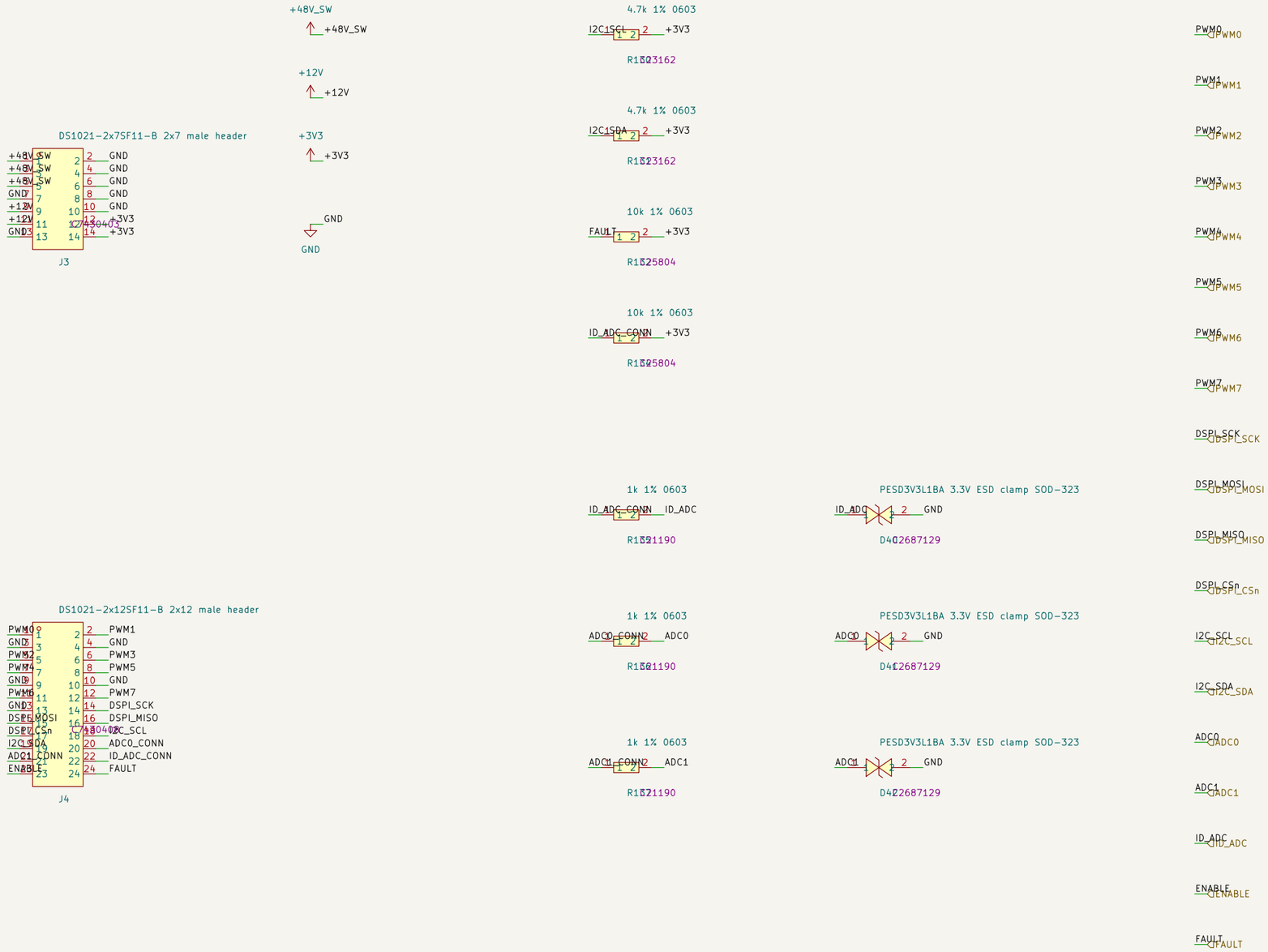
J2 SILKSCREEN: "PoE OFF or ISOLATED ADAPTER ONLY" — an earthed USB-UART adapter breaks PD signature detection

GPIO45 (U30 pad 26) IS A NO-CONNECT BY DESIGN: pulled high at power-up on this no-PSRAM -N8 it selects 1.8 V VDD_SPI and the 3.3 V flash never boots

BOOT idles HIGH through R101 (GPIO0 = 1 -> SPI Boot): SW1 and J2 pin 6 can only pull it low

Keep the antenna end of U30 (the pad 1 / pad 40 corner) clear of copper, parts and traces on EVERY layer

ai-ee		
Sheet: /mcu/		
File: mcu.kicad_sch		
Title: LUM-CAR-A: ESP32-S3 module, EN/BOOT, recovery header		
Size: A3	Date: 2026-07-28	Rev: 1
KiCad E.D.A. 10.0.3		Id: 2/6



ai-ee

Sheet: /expansion/

File: expansion.kicad_sch

Title: LUM-CAR-A: expansion - J3 power / J4 signal (ICD-01 rev A2)

Size: A3	Date: 2026-07-28	Rev: 1
KiCad E.D.A. 10.0.3		Id: 2/6